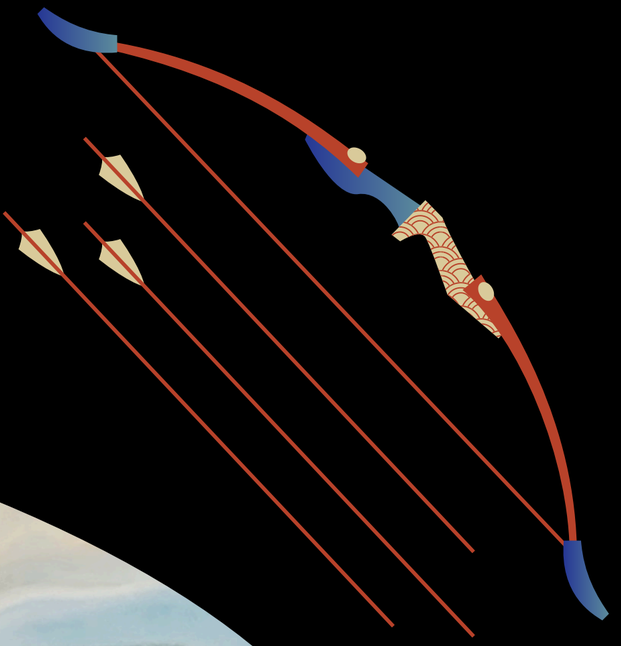
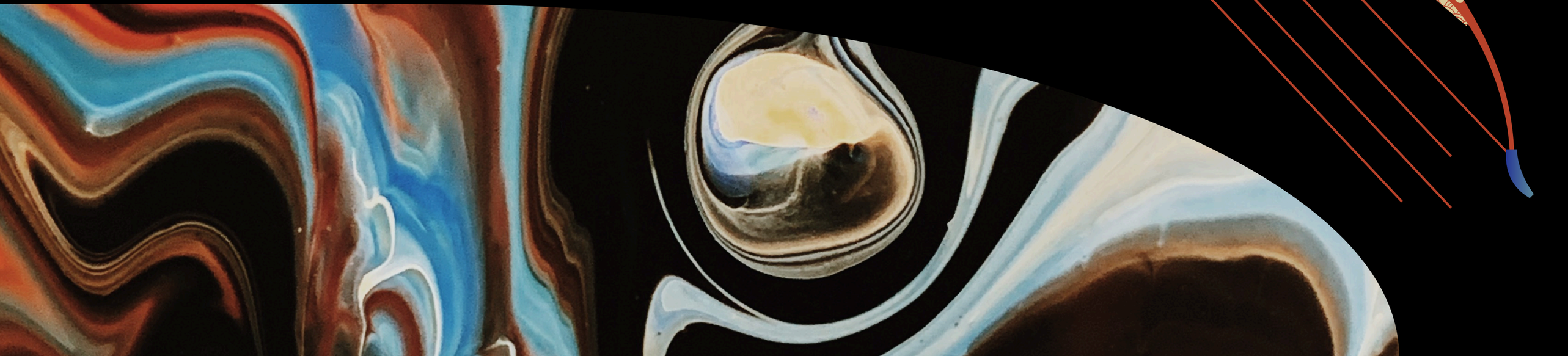


ARTEMIS

USC Makers | Sohana Singh

or, Archery Ranging and Testing Equipment for Mastery of Immersive Skills



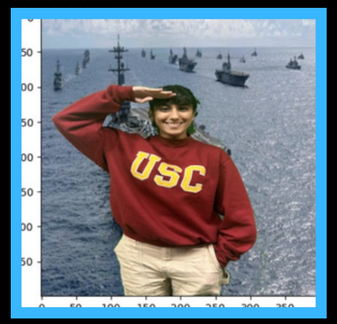


ABOUT ME

- Sophomore in ECE
- In Makers since Fall 2025
- Hometown: Dallas, TX
- Also in the USC Rocket Propulsion Laboratory and the Student Symphony Orchestra
- Hobbies / Interests: Video games (Zelda, Final Fantasy, etc), Photography, Skateboarding, and anything nature :)

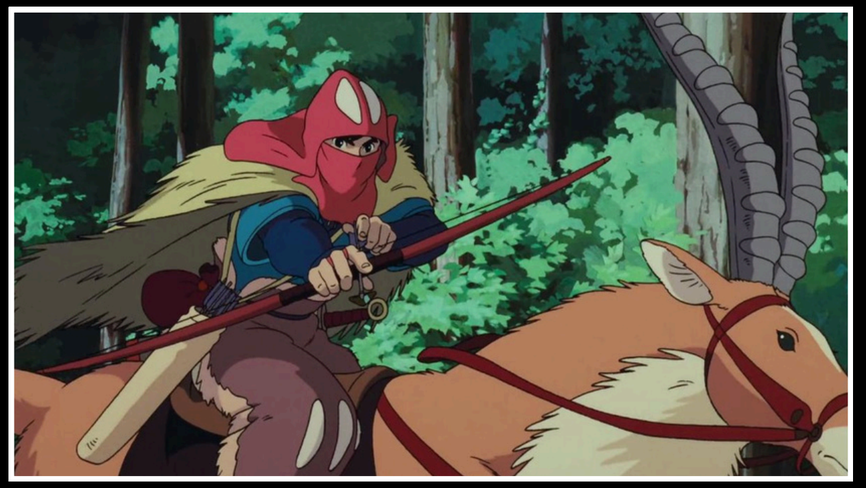


credit to reina











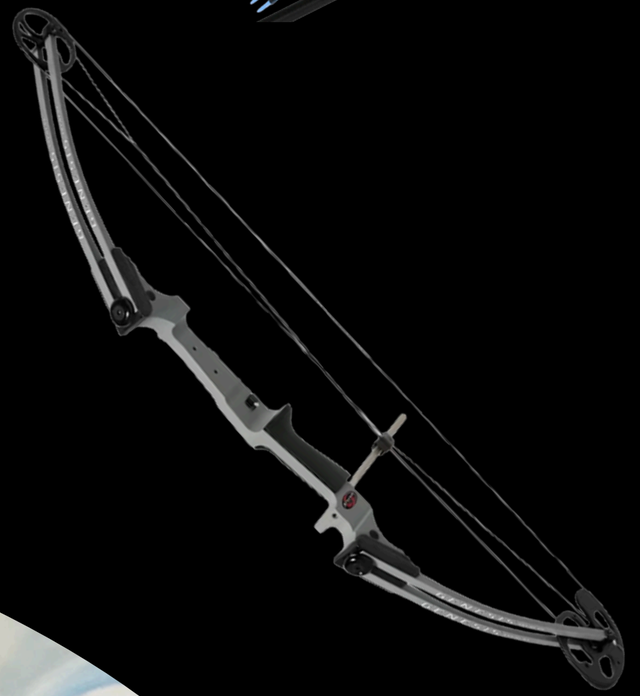


INSPIRATION

WHAT IS ARTEMIS?

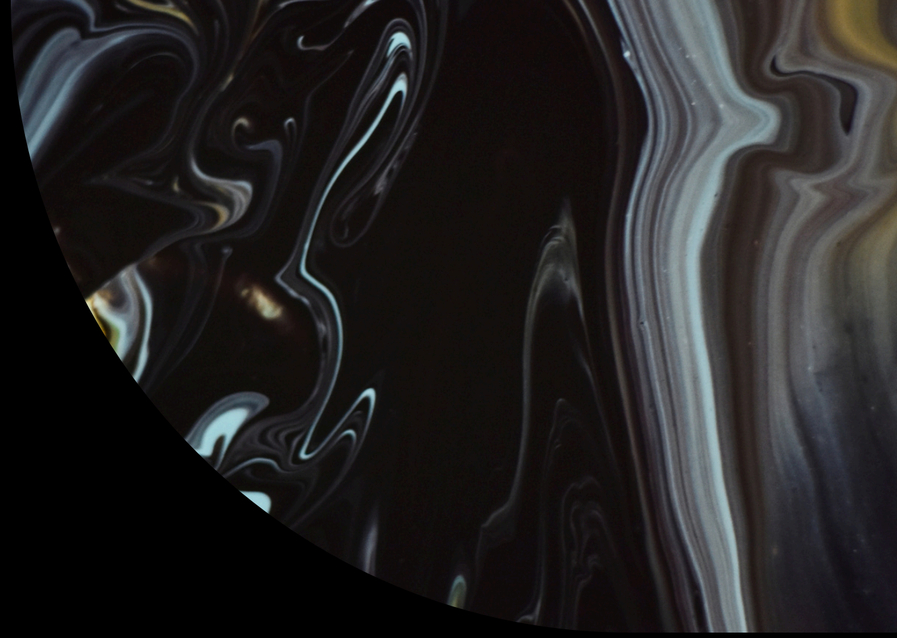
- Physical bow-and-arrow set
- Aim assistant – guides the user via auditory or visual commands on how to obtain the perfect bullseye
    - Pitch/angle of bow, orientation/yaw of bow, and draw of arrow / strength of the shot taken into account to provide recommended parameters







SUBTEAMS



ELECTRICAL / SENSING:

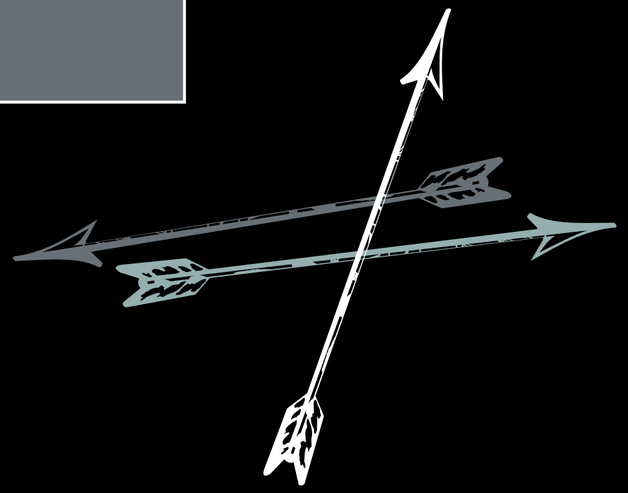
Work on choosing and implementing sensors (camera, gyroscope, tension sensing, etc.), and integrating with microcontroller and power systems

SOFTWARE:

Develop physics simulation for arrow path, and algorithm for determining corrections to user state recorded by on-bow sensors and environmental target factors

MECHANICAL:

Work on physical bow-and-arrow design, making the bow from wood or using pre-made materials*, and integrating physical electronic systems seamlessly and ergonomically with the frame



FALL SEMESTER

- Finished physical bow and arrow frame*
- Sensor configuration determined and sensor hardware developed
- Software physics simulations developed

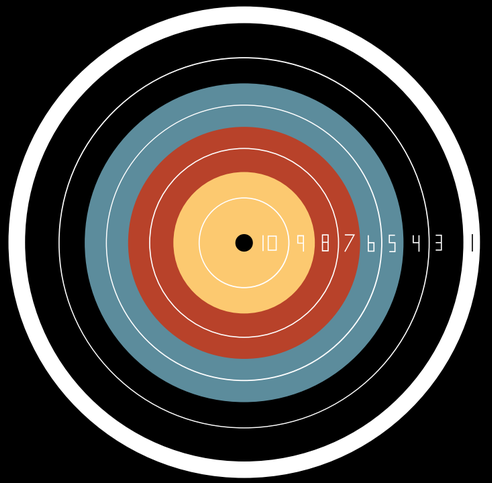


SPRING SEMESTER

- Integrate sensors with physical frame (harnessing)
- Develop sensor fusion systems and intregrate sensors with software
- Increase precision of physics simulations and arrow path predictions

stretch goal: be able to accurately shoot a moving target!

*notes on SCOPE:



Physical bow and arrow:

Do we make it ourselves out of wood, or do we buy and modified a pre-made one?

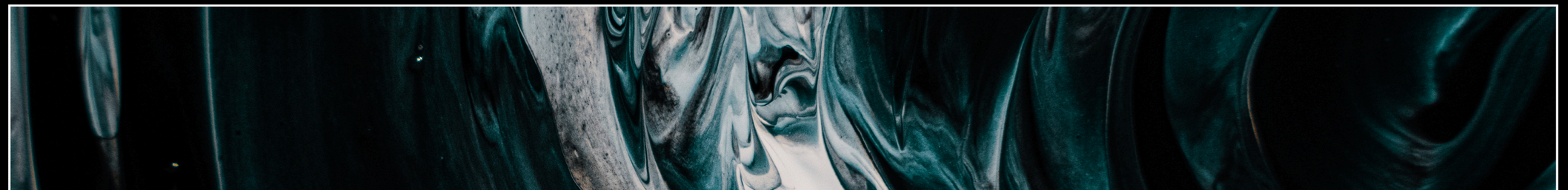
Pros of making it:

- We control more parameters
- Easier for software to account for and easier for sensors to be designed around
- It is very cool and great experience

Pros of buying it:

- We can be sure it works well
- Limits points of failure
- Saves time

Full team will make final decisions!!





WHY JOIN?

- Very interdisciplinary
- Challenging project, and many opportunities to learn
- You'll have ownership over the direction and execution of the project
- You get to learn a bit of archery!



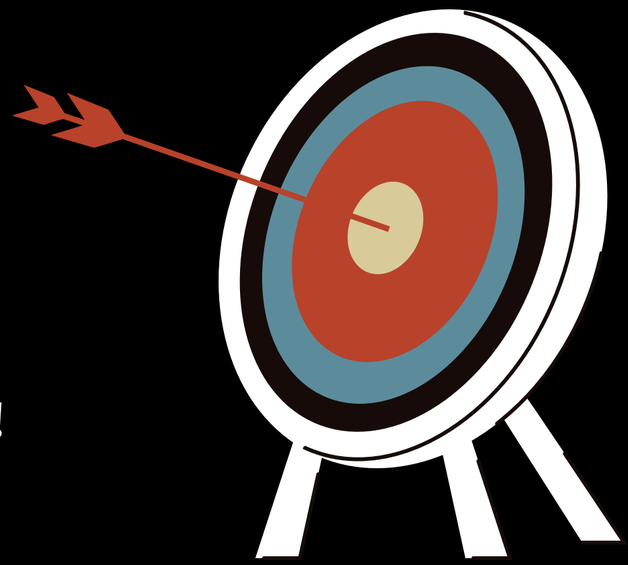
What I will provide:

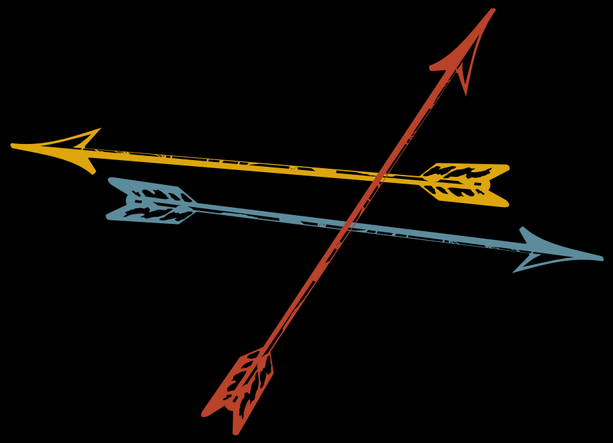
- Support – I'll help you directly as much as I can, and help find resources if there's something new!
- Team bonding and a good community!





THANK YOU!





Reach out if you have any questions!

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